Andhra University College of Engineering

Mechanical Engineering | Course Code: ME301

Engineering Thermodynamics - Comprehensive Syllabus Lecture Notes

Syllabus Focus: ME301 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Engineering Thermodynamics. Includes detailed

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Zeroth & First Law of Thermodynamics

" Syllabus Scope: Thermodynamic systems, Work and heat interactions, First law for closed and steady-flow

Unit 2: Second Law of Thermodynamics & Entropy

" Syllabus Scope: Kelvin-Planck and Clausius statements, Reversibility, Carnot cycle, Clausius inequality

ME301: Engineering Thermodynamics - Advanced Topics & Assessment

Academic Track: Mechanical Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Properties of Pure Substances & Steam

" P-v-T surfaces, Phase change processes, Mollier diagram, Steam tables, Dryness fraction measurement, Throttl

Unit 4: Vapor Power Cycles Rankine

" Ideal Rankine cycle, Thermal efficiency, Mean temperature of heat addition, Reheat and Regenerative Rankine

Unit 5: Gas Power Cycles

" Air standard cycles: Otto, Diesel, Dual combustion cycles, Brayton cycle for gas turbines, Intercooling, Reh

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Derive the thermal efficiency of an air-standard Diesel cycle in terms of compression ratio and cut-of
2. Formulate the steady-flow energy equation SFEE and apply it to a steam turbine and converging-divergin
3. Analyze an ideal Regenerative Rankine cycle with one open feedwater heater and derive its thermal effi

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Yunus A. Cengel, Michael A. Boles - Thermodynamics: An Engineering Approach McGraw-Hill
* P.K. Nag - Engineering Thermodynamics Tata McGraw-Hill 6th Ed

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